Automobiles Dataset Analysis

Master MLDM – Data Analysis Project

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1 Introduction

In this project, we analyze the *Automobiles* dataset, which contains technical characteristics of cars from this period. Our objectives are to:

- describe and explore the dataset;
- study the statistical properties of the variables;
- model fuel efficiency using several regression methods;
- apply dimensionality reduction (PCA) and use it both for visualization and as a preprocessing step for regression;
- explore potential structures through clustering.

All analyses were conducted in Python using only the authorized libraries (`NumPy`, `Pandas`, `Matplotlib`, `Seaborn`, `SciPy`, `Scikit-learn`).

2 Dataset Description

The dataset contains information on automobiles produced between the early 1970s and the early 1980s. Each row corresponds to a single vehicle, described by several technical and contextual variables.

2.1 Variables

The main variables are:

- `miles_per_gallon`: fuel efficiency (continuous);
- `cylinders`: number of engine cylinders (discrete);
- `displacement`: engine displacement in cubic inches (continuous);
- `horsepower`: engine power (continuous);
- `weight_lbs`: vehicle weight in pounds (continuous);
- `acceleration`: time to accelerate from 0 to 60 mph (continuous);
- `model_year`: year of production (discrete, from 1970 to 1982);
- `origin_country`: encoded origin of the car (1 = USA, 2 = Europe, 3 = Japan).

The dataset contains 392 complete observations, with no missing values after preprocessing.

2.2 Descriptive Statistics

Table 1 reports the descriptive statistics for the quantitative variables.

	MPG	Cylinders	Displacement	Horsepower	Weight (lbs)	Acceleration	Model Year
Count	392	392	392	392	392	392	392
Mean	23.45	5.47	194.41	104.47	2977.58	15.54	75.98
Std	7.81	1.71	104.64	38.49	849.40	2.76	3.68
Min	9.0	3	68	46	1613	8.0	70
25%	17.0	4	105	75	2225	13.78	73
Median	22.75	4	151	93.5	2803.5	15.5	76
75%	29.0	8	275.75	126	3614.75	17.03	79
Max	46.6	8	455	230	5140	24.8	82

Table 1: Descriptive statistics of the quantitative variables

Fuel efficiency (`miles_per_gallon`) varies substantially across vehicles, from 9 to 46.6 MPG, with a mean around 23.5 MPG. Engine-related variables such as `displacement`, `horsepower` and `cylinders` show wide ranges, reflecting very different engine sizes and power levels. Vehicle weight spans from about 1600 to more than 5000 pounds, indicating the presence of both relatively light and very heavy cars. The acceleration time also varies considerably, and the `model_year` variable covers the period from 1970 to 1982.

Several variables (e.g., `displacement`, `horsepower`, `weight_lbs`) exhibit wide ranges and are likely positively skewed. A logarithmic transformation could potentially reduce skewness and stabilize variance; however, in this work we choose not to apply log-transforms in order to maintain direct interpretability of the original scale.

2.3 Fuel Efficiency Across Origins

We also compare fuel efficiency across different regions of origin (USA, Europe, Japan). Average MPG values are reported in Table 2.

Origin	Mean MPG	CAFE (24 MPG)
USA	20.03	< 24 (does not meet)
Europe	27.60	≥ 24
Japan	30.45	≥ 24

Table 2: Average fuel efficiency by origin (USA, Europe, Japan)

Average fuel efficiency is lowest for cars manufactured in the USA (around 20 MPG), intermediate for European cars (about 27.6 MPG), and highest for Japanese cars (about 30.5 MPG). The CAFE standard requires an average fuel efficiency of at least 24 MPG. According to this dataset, cars originating from the USA do not reach this threshold on average, whereas European and Japanese cars clearly exceed it.

3 Exploratory Data Analysis

3.1 Pairwise Relationships

Pairwise relationships between fuel efficiency and the main technical variables were explored using scatter plots and a pairplot.

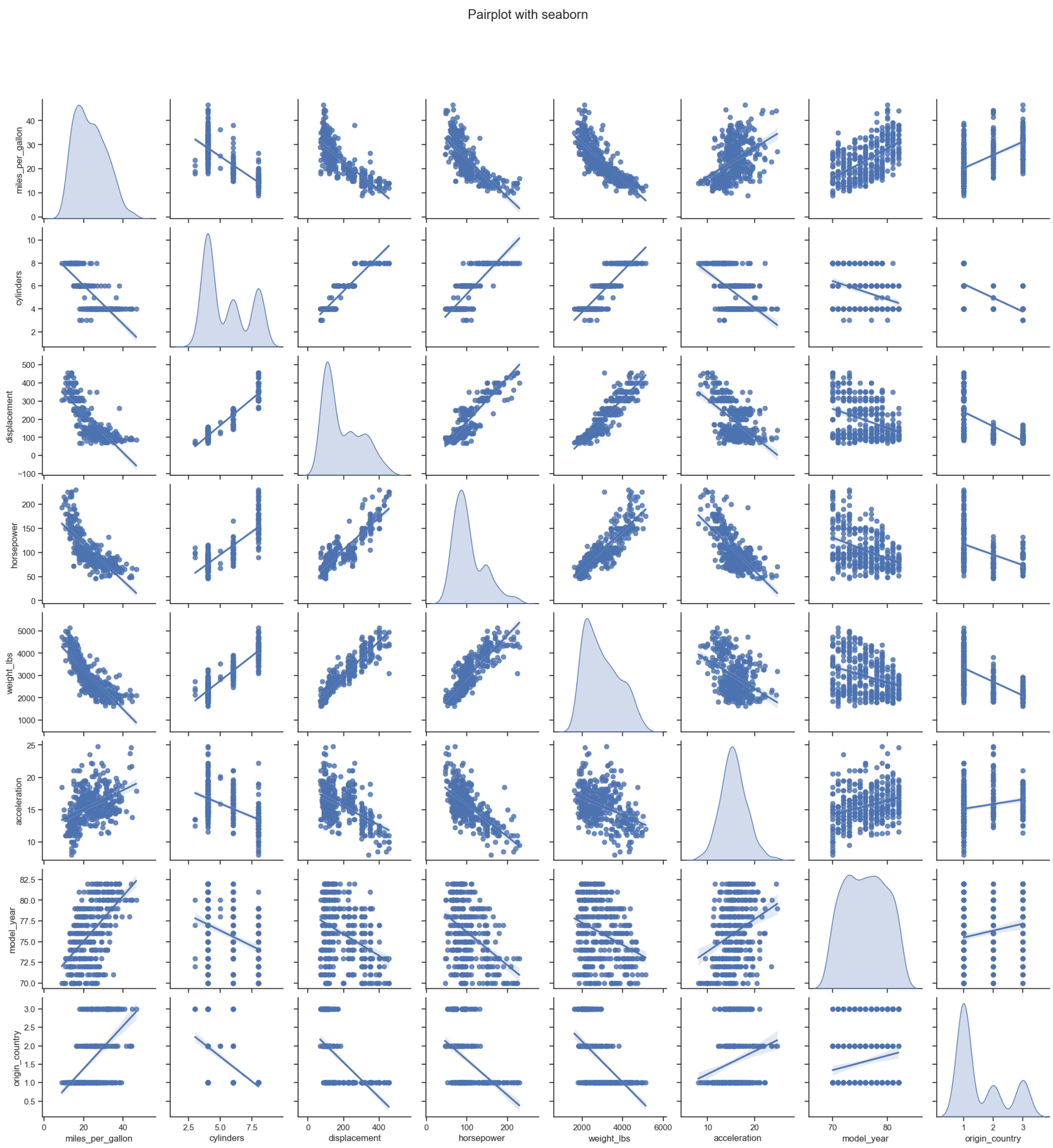


Figure 1: Pairplot of selected variables (including MPG and main engine characteristics)

The pairplot visually suggests strong negative relationships between `miles_per_gallon` and variables such as `horsepower`, `weight_lbs` and `displacement`: cars that are heavier or more powerful tend to consume more fuel (lower MPG). Fuel efficiency appears positively associated with `model_year` and `origin_country`, suggesting that more recent models and certain origins tend to be more efficient. Some potential outliers can be seen in these plots, but since the bulk of the distributions remains reasonably close to Gaussian shapes, we keep all observations in the present analysis.

3.2 Correlation Matrix

To quantify linear relationships, we compute the correlation matrix of the quantitative variables. The empirical correlations are as follows (values used for interpretation):

- `miles_per_gallon` is strongly negatively correlated with `cylinders` (-0.78), `displacement` (-0.81), `horsepower` (-0.78) and `weight_lbs` (-0.83), and positively correlated with `acceleration` (0.42), `model_year` (0.58) and `origin_country` (0.57).

- Among the predictors themselves, there are very high positive correlations: `cylinders` and `displacement` (0.95), `displacement` and `weight_lbs` (0.93), `cylinders` and `weight_lbs` (0.90), `displacement` and `horsepower` (0.90), etc.
- `acceleration` tends to be negatively correlated with engine size and power (correlations between -0.54 and -0.69), and positively correlated with `miles_per_gallon` and `model_year`.

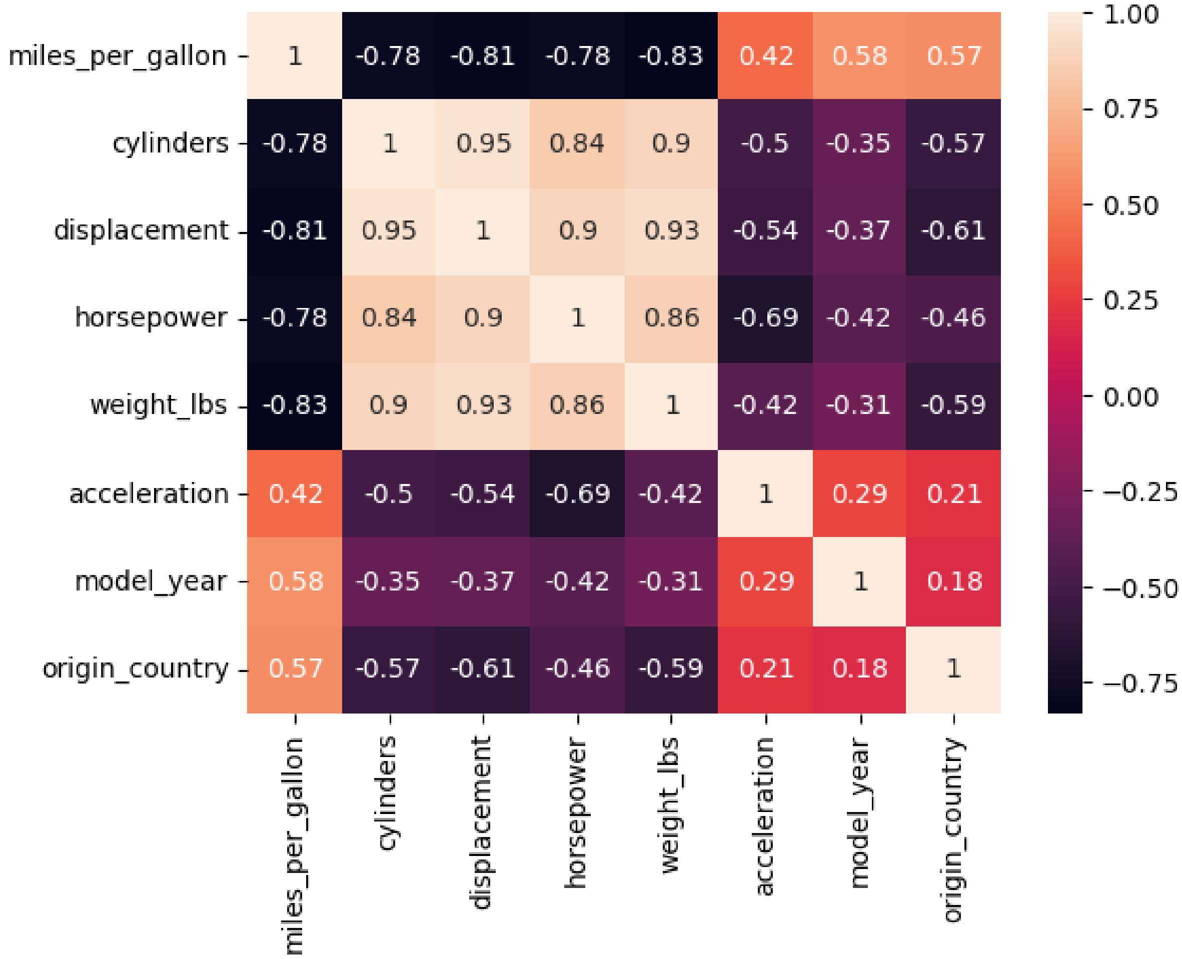


Figure 2: Correlation matrix of the quantitative variables

These values confirm that fuel efficiency is strongly related to engine size and vehicle weight, and also to model year and origin. They also indicate substantial redundancy among some features (e.g., `cylinders`, `displacement`, `weight_lbs` and `horsepower` are all strongly correlated), which motivates the later use of PCA to reduce dimensionality by combining these variables into a smaller number of components.

4 Normality Assessment

Normality of the quantitative variables was assessed using both graphical and statistical tools.

4.1 Q–Q Plots of Variables

Q–Q plots were generated for each quantitative variable, comparing empirical quantiles to the theoretical quantiles of a normal distribution.

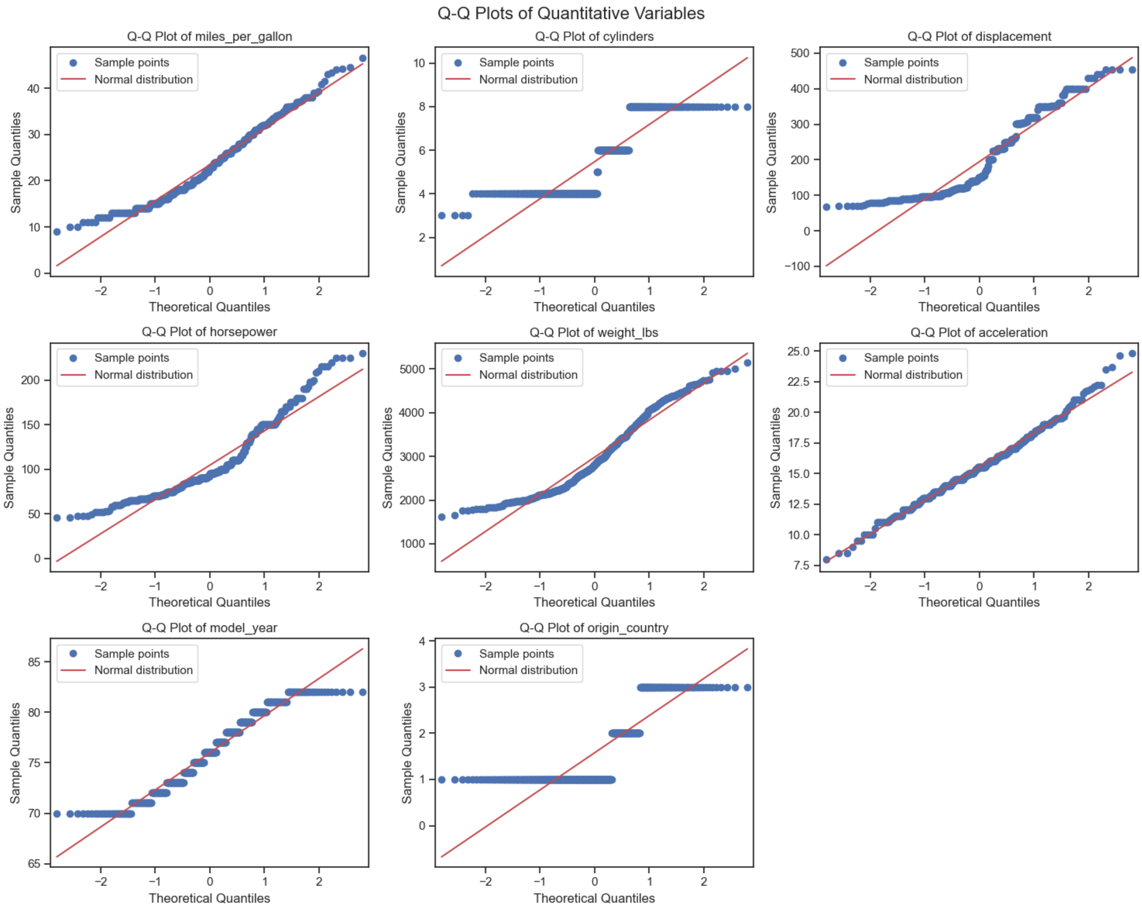


Figure 3: Q–Q plots of the quantitative variables

For most variables, the Q–Q plots show clear deviations from the straight reference line, especially in the tails. Engine-related variables and weight tend to exhibit heavy tails and asymmetry, while discrete variables such as `cylinders`, `model_year` and `origin_country` naturally produce step-like patterns rather than smooth Gaussian shapes. Among the variables, `acceleration` appears visually closest to a normal distribution, although some deviations remain visible in the extremes.

4.2 Shapiro–Wilk Test

The Shapiro–Wilk test was applied to each quantitative variable. Table 3 summarizes the test statistics and p-values.

Variable	Shapiro statistic	p-value
miles_per_gallon	0.9672	1.05×10^{-7}
cylinders	0.7507	6.88×10^{-24}
displacement	0.8818	8.98×10^{-17}
horsepower	0.9041	5.02×10^{-15}
weight_lbs	0.9415	2.60×10^{-11}
acceleration	0.9919	3.05×10^{-2}
model_year	0.9470	1.22×10^{-10}
origin_country	0.6738	8.80×10^{-27}

Table 3: Shapiro–Wilk test for the quantitative variables

The very small p-values for most variables (often much smaller than 10^{-7}) indicate strong departures from normality, which is consistent with the pronounced curvature and tail deviations observed in the Q–Q plots. `Acceleration` has the largest Shapiro p-value (approximately 0.03) and a statistic close to 1, which is in line with its Q–Q plot being relatively close to the straight line, although some deviations persist. Discrete variables such as `cylinders` and `origin_country` naturally show very strong departures from a Gaussian shape in both the Q–Q plots and the Shapiro statistics.

These findings show that the original variables are not exactly Gaussian, although some of them (in particular `acceleration` and `miles_per_gallon`) are roughly normal if we accept a coarse Gaussian approximation. This is not surprising in this context and does not prevent us from fitting regression models, since the normality assumption is more important for the residuals than for the raw variables themselves.

5 Univariate Regression: MPG vs Horsepower

Given the strong empirical correlation between `miles_per_gallon` and `horsepower`, we first study a univariate regression model where fuel efficiency is expressed as a function of engine power.

5.1 Modeling Pipeline

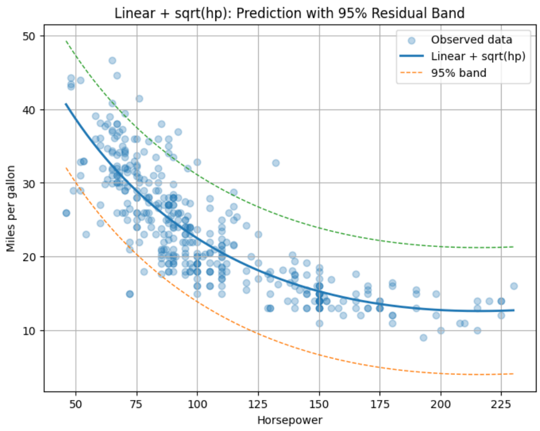
We consider three regression approaches using only `horsepower` as explanatory variable:

- a linear regression model with a transformation of the input, using both `horsepower` and $\sqrt{\texttt{horsepower}}$ as features;
- a Ridge regression model;
- a multilayer perceptron (MLP) regressor.

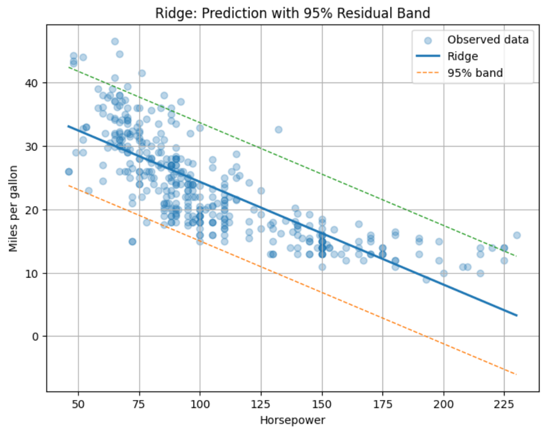
In all cases, the data are split into training and test sets. Standardization is applied when appropriate (Ridge and MLP) via a `Pipeline`. The linear model with $\sqrt{\texttt{horsepower}}$ allows some non-linearity while remaining interpretable, whereas the MLP can approximate more flexible relationships.

5.2 Regression Curves and Confidence Intervals

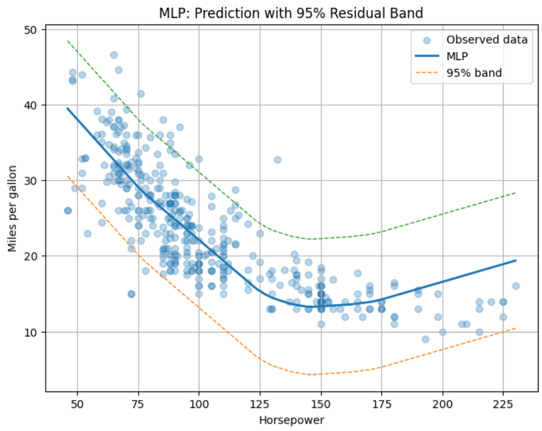
Figure 4 shows the learned relationships between horsepower and fuel efficiency for the three models. For the linear model, a 95% confidence interval around the predicted mean is displayed. Under the usual Gaussian assumption on the errors, this interval is constructed as $\hat{y} \pm 1.96 \times \widehat{\sigma}$, where $\widehat{\sigma}$ is an estimate of the residual standard deviation. The factor 1.96 corresponds to the approximate 97.5% quantile of the standard normal distribution, so that about 95% of the mass lies within ± 1.96 standard deviations of the mean.



(a) Linear regression + $\sqrt{\texttt{hp}}$



(b) Ridge regression



(c) MLP regression

Figure 4: Univariate models of MPG as a function of horsepower

All three models reproduce the expected decreasing relationship: higher horsepower is associated with lower fuel efficiency, with some curvature captured more flexibly by the linear+$\sqrt{\texttt{hp}}$ and MLP models.

5.3 Performance Metrics

Model performance is evaluated on the test set using RMSE and R^2. Table 4 summarizes the results.

Model	RMSE	R^2
Linear + $\sqrt{\texttt{horsepower}}$	18.81	0.63
Ridge (univariate)	22.11	0.57
MLP (univariate)	20.32	0.60

Table 4: Performance of univariate models (MPG vs horsepower)

In this univariate setting, all three models explain a moderate proportion of the variance in fuel efficiency, with R^2 values between 0.57 and 0.63. The best-performing model according to RMSE and R^2 is the linear model with the additional $\sqrt{\texttt{horsepower}}$ feature, followed by the MLP and then the univariate Ridge regression. This suggests that a simple non-linear transformation of horsepower already captures a substantial part of the relationship without requiring a highly complex model.

5.4 Residual Diagnostics

To assess the validity of the modeling assumptions, we examine the residuals. Figure 5 displays Q–Q plots of the residuals, and Table 5 reports Shapiro–Wilk test results.

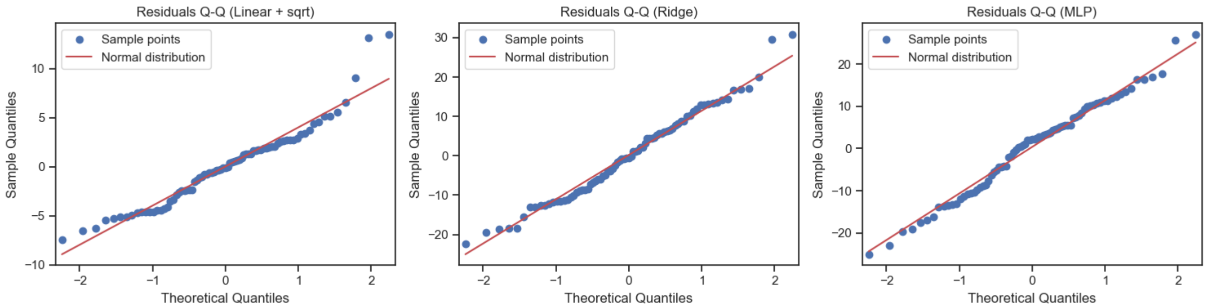


Figure 5: Q–Q plots of residuals for univariate models

Model	Shapiro statistic	p-value
Linear + $\sqrt{\texttt{horsepower}}$	0.9752	0.1267
Ridge (univariate)	0.951021	0.004241
MLP (univariate)	0.989284	0.756490

Table 5: Shapiro–Wilk test on residuals of univariate models

For the linear model with $\sqrt{\texttt{horsepower}}$, the Q–Q plot is reasonably close to the diagonal, with moderate deviations in the tails, and the Shapiro statistic (0.9752) together with a p-value around 0.13 suggest residuals that are not strongly incompatible with a normal shape. For the univariate Ridge model, the Q–Q plot shows more pronounced deviations and the p-value is very small (0.0042), indicating residuals that deviate more clearly from normality. In contrast, the MLP model yields residuals with a Q–Q plot very close to the straight line and a large Shapiro p-value (0.7565), which is consistent with residuals behaving in a way that is much more similar to a Gaussian distribution.

Overall, in this univariate context, the linear+$\sqrt{\texttt{hp}}$ and especially the MLP model exhibit residuals whose distribution is reasonably close to normal, while the univariate Ridge model shows stronger departures.

6 Multivariate Regression

We now consider a multivariate setting where several vehicle characteristics are used simultaneously to predict fuel efficiency.

6.1 Feature Set and Pipeline

The following variables are included as predictors:

- `horsepower`,
- `weight_lbs`,
- `displacement`,
- `cylinders`,

- `acceleration`,
- `model_year`,
- `origin_country`.

Two models are considered:

- Ridge regression with standardized features,
- MLP regression with standardized features.

The models are implemented using `Pipeline` objects that include a `StandardScaler` followed by the chosen regressor.

6.2 Performance and Residuals

Table 6 reports the RMSE and R^2 on the test set for both multivariate models.

Model	RMSE	R^2
Ridge (multivariate)	10.77	0.79
MLP (multivariate)	9.12	0.82

Table 6: Performance of multivariate regression models

Compared to the univariate models, multivariate regression substantially improves predictive performance: RMSE decreases (down to about 9–11 MPG) and R^2 increases to 0.79 for Ridge and 0.82 for the MLP. This confirms that additional variables such as weight, displacement, cylinders, model year and origin provide complementary information beyond horsepower alone. The MLP again slightly outperforms Ridge, likely due to its ability to capture non-linear interactions between features.

Residual diagnostics are again performed using Q–Q plots and Shapiro–Wilk tests (Figure 6, Table 7).

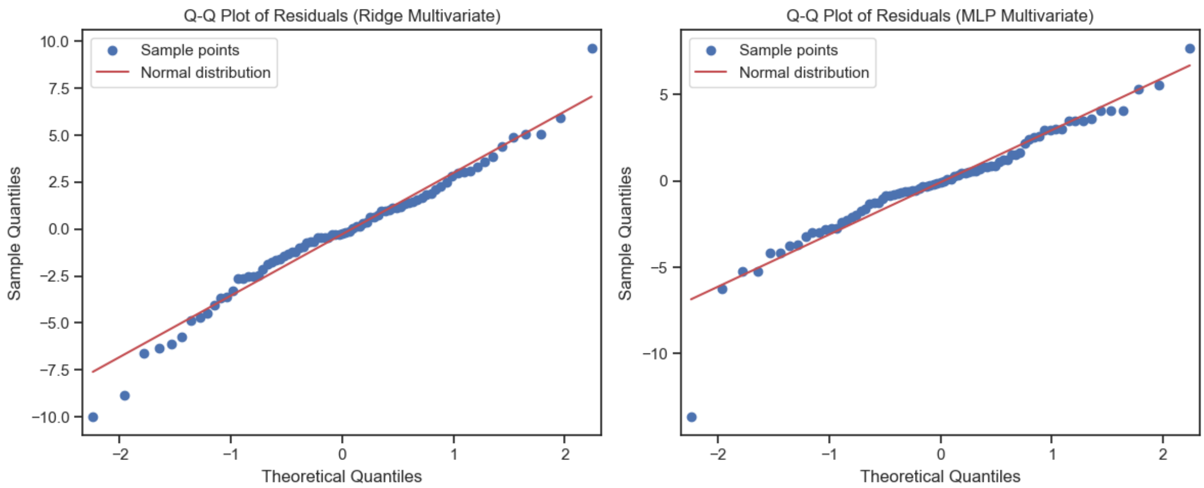


Figure 6: Q–Q plots of residuals for multivariate Ridge and MLP models

Model	Shapiro statistic	p-value
Ridge (multivariate)	0.9795	0.2344
MLP (multivariate)	0.9403	0.0010

Table 7: Shapiro–Wilk test on residuals (multivariate models)

For the multivariate Ridge model, the Q–Q plot of the residuals is close to the diagonal, and the Shapiro statistic (0.9795) with p-value around 0.23 suggests residuals that are reasonably compatible with a normal distribution. For the multivariate MLP, the Q–Q plot shows more pronounced deviations in the tails, and the p-value is very small (0.0010), indicating that the residuals depart more clearly from normality. This illustrates that a model can achieve slightly better predictive performance (lower RMSE, higher R^2) while producing residuals that are less Gaussian.

Finally, Figure 7 shows the observed versus predicted MPG values for both models, providing a visual assessment of the fit quality.

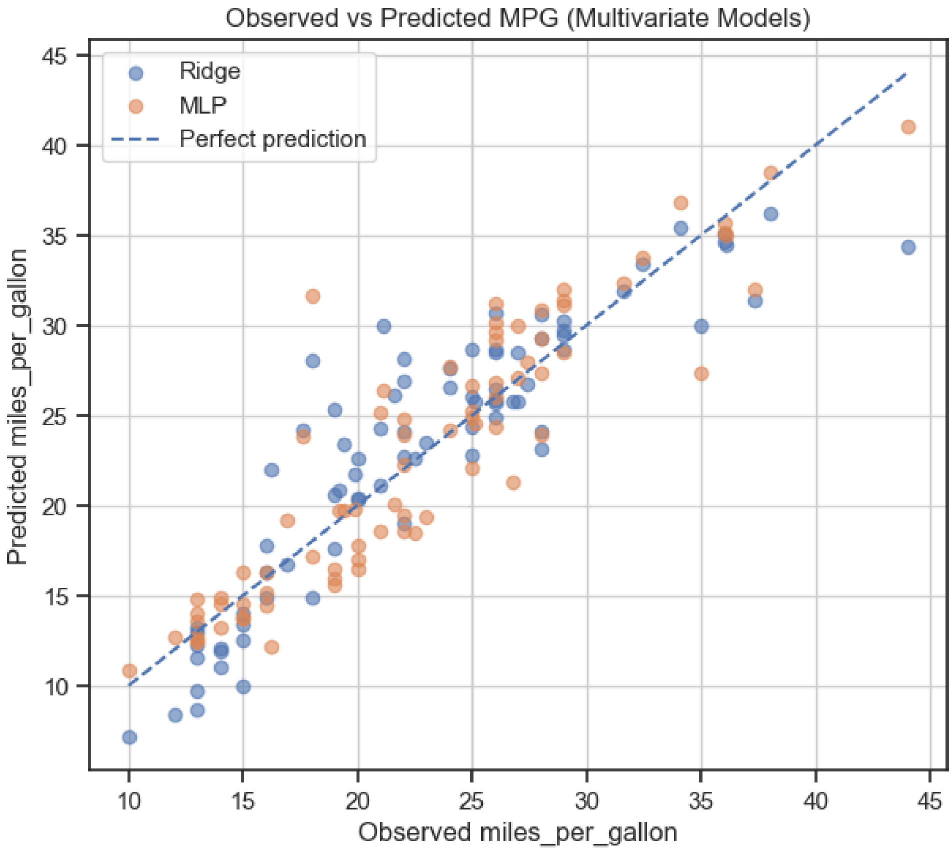


Figure 7: Observed vs predicted MPG for multivariate Ridge and MLP models

Both models produce points that are reasonably close to the diagonal line, especially the MLP, which is consistent with the higher R^2 observed in Table 6.

7 Principal Component Analysis (PCA)

7.1 PCA on Standardized Features

PCA is applied to the standardized explanatory variables (`horsepower`, `weight_lbs`, `displacement`, `cylinders`, `acceleration`, `model_year`, `origin_country`). Figure 8 shows the explained variance ratio and its cumulative sum.

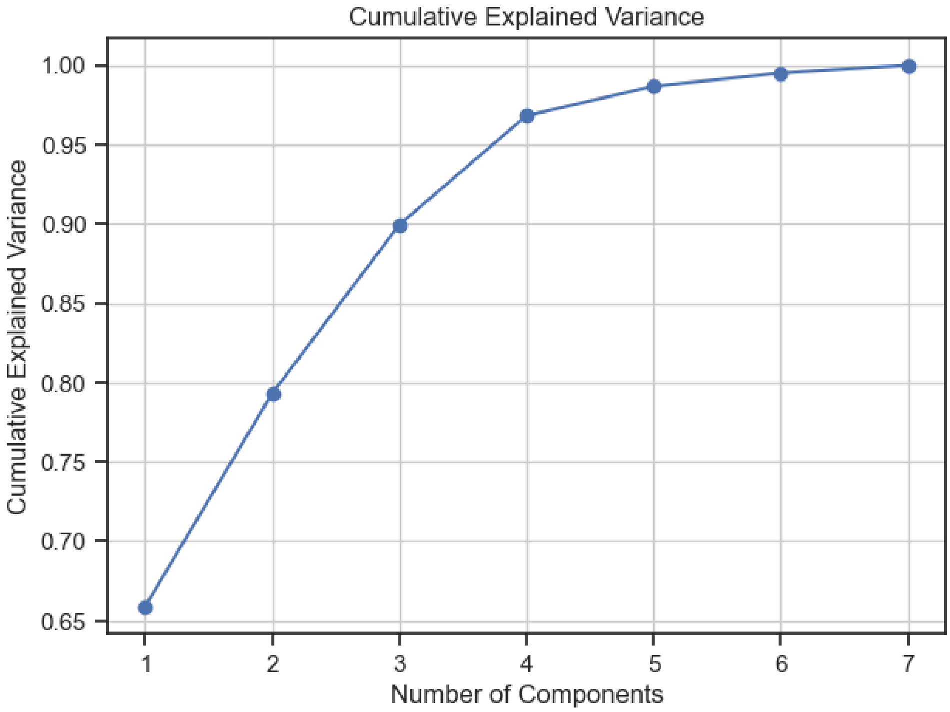


Figure 8: Explained variance and cumulative explained variance of PCA

The explained variance ratios for the components are as follows: PC1 explains about 65.9% of the total variance, PC2 adds 13.4% (cumulative 79.3%), PC3 adds 10.6% (cumulative 89.9%), PC4 adds 6.9% (cumulative 96.8%), PC5 adds 1.9%, PC6 about 0.8% and PC7 about 0.5%. Thus, the first three principal components together capture almost 90% of the variance in the standardized features, while the remaining components contribute relatively little additional variance.

Due to the strong correlations observed in the correlation matrix (Figure 2), many variables carry redundant information. The PCA loadings (not shown here) indicate that the first principal component is mainly driven by engine size and power variables (`displacement`, `horsepower`, `weight_lbs` and `cylinders`), whereas the second and third components are more related to `model_year`, `acceleration` and `origin_country`. We therefore choose to retain the first three principal components, which capture a large proportion of the total variance and summarize the main underlying patterns.

7.2 Regression on PCA Components

We build new multivariate models using only the first three principal components as inputs. As before, we consider:

- Ridge regression on the three PCs;
- MLP regression on the three PCs.

Table 8 summarizes the RMSE and R^2 for these PCA-based models.

Model	RMSE	R^2
Ridge on 3 PCs	13.00	0.75
MLP on 3 PCs	8.18	0.84

Table 8: Performance of regression models using the first 3 principal components

Using only three components leads to a clear loss of performance for Ridge compared to the full multivariate model: the RMSE increases from 10.77 to 13.00 and R^2 drops from 0.79 to 0.75. This is expected, since projecting the predictors onto a lower-dimensional space inevitably discards some information.

Interestingly, the MLP model behaves differently. On the three principal components, it reaches an RMSE of 8.18 and an R^2 of 0.84, which is slightly better than the MLP trained on the original variables. Here, PCA seems to help the non-linear model by providing cleaner, orthogonal inputs and removing redundant directions that do not bring useful signal for prediction.

Additional experiments (not reported in tables) show that increasing the number of components beyond three does not improve predictive performance and can even slightly degrade it. This is consistent with the fact that later principal components explain very little variance and are often dominated by noise or by directions weakly related to the target variable `miles_per_gallon`. Including such components increases model complexity without adding meaningful signal, which can hurt generalization.

Residual diagnostics (Q–Q plots and Shapiro–Wilk tests) are also performed (Figure 9, Table 9).

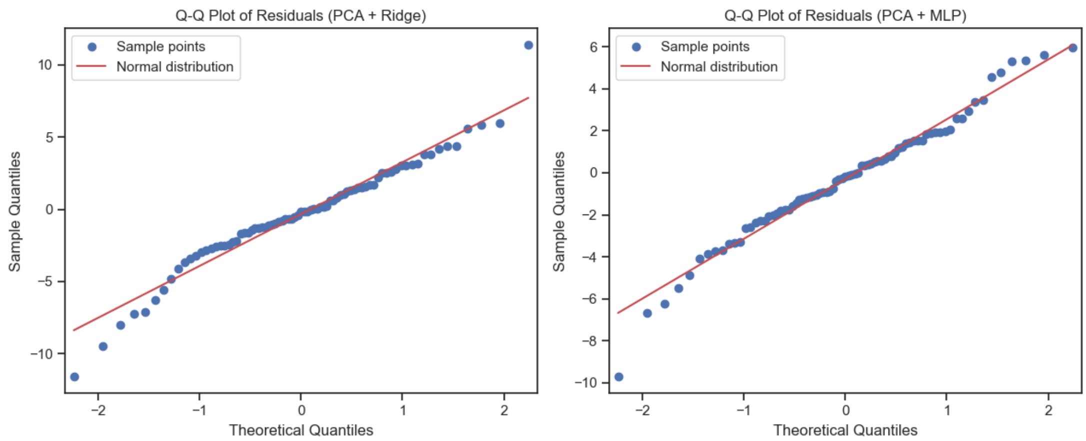


Figure 9: Q–Q plots of residuals for PCA-based regression models

Model	Shapiro statistic	p-value
Ridge on 3 PCs	0.9686	0.04890
MLP on 3 PCs	0.9810	0.2899

Table 9: Shapiro–Wilk test on residuals (PCA-based models)

For Ridge on three PCs, the Q–Q plot shows some deviations in the tails, and the Shapiro statistic (0.9686) with p-value around 0.049 indicates residuals that are slightly further from normality than in the full multivariate Ridge model. For the MLP on three PCs, the Q–Q plot is closer to the diagonal and the Shapiro p-value is relatively large (0.2899), suggesting residuals that are more compatible with a normal distribution. Thus, PCA as a preprocessing step tends to improve the residual behavior of the MLP, while slightly deteriorating that of Ridge compared to using the original variables.

8 Global Comparison of Models

To summarize, Table 10 compares all models in terms of RMSE, R^2 and residual normality (based on the Shapiro–Wilk test).

Model	RMSE	R^2	Residual normality (Shapiro)
Linear + $\sqrt{\texttt{horsepower}}$	18.81	0.63	Residuals reasonably close to normal (p = 0.13)
Ridge (univariate)	22.11	0.57	Clear deviation from normality (p = 0.004)
MLP (univariate)	20.32	0.60	Residuals closest to normal (p = 0.76)
Ridge (multivariate)	10.77	0.79	Residuals reasonably close to normal (p = 0.23)
MLP (multivariate)	9.12	0.82	Residuals with marked tail deviations (p = 0.001)
Ridge on 3 PCs	13.00	0.75	Slight deviation from normality (p = 0.049)
MLP on 3 PCs	8.18	0.84	Residuals reasonably close to normal (p = 0.29)

Table 10: Global summary of model performances and residual normality

From Tables 4, 6 and 8, we observe a clear progression from univariate to multivariate models, and, in some cases, an additional benefit when using PCA as a preprocessing step. Univariate models achieve moderate R^2 values around 0.6, whereas multivariate Ridge and MLP models reach R^2 values close to 0.8. Using three principal components leads to slightly worse performance for Ridge (higher RMSE and lower R^2 than the full multivariate Ridge model), which reflects the loss of information when projecting the predictors onto a lower-dimensional space. In contrast, the MLP trained on three PCs attains the best overall predictive performance (lowest RMSE and highest R^2), suggesting that the orthogonalized PCA representation is particularly beneficial for this non-linear model.

Regarding residual behavior, the Shapiro–Wilk tests and the corresponding Q–Q plots show that:

- the univariate MLP and the MLP on three PCs produce residuals that are closest to a normal distribution, with Q–Q plots very close to the diagonal and large Shapiro p-values;
- the linear model with $\sqrt{\texttt{horsepower}}$ and the multivariate Ridge model yield residuals that are reasonably close to normal, with mild deviations mainly in the tails;
- the univariate Ridge model and the multivariate MLP without PCA exhibit more pronounced deviations from normality, especially in the extremes of the Q–Q plots, which is reflected in their very small p-values;
- the Ridge model on three PCs lies in between, with slight tail deviations and a Shapiro statistic indicating a modest departure from normality.

Overall, this comparison highlights a trade-off between model complexity, predictive performance and residual behavior: multivariate and PCA-based models generally perform better in terms of RMSE and R^2, while the degree to which residuals resemble a normal distribution depends on both the choice of model (linear vs MLP) and the representation of the inputs (original features vs principal components).

9 Clustering Analysis

9.1 K-means Clustering in PCA Space

Unsupervised clustering is performed using K-means with $k = 3$ clusters on the standardized original features. The resulting cluster assignments are then visualized in the two-dimensional PCA space spanned by the first two principal components.

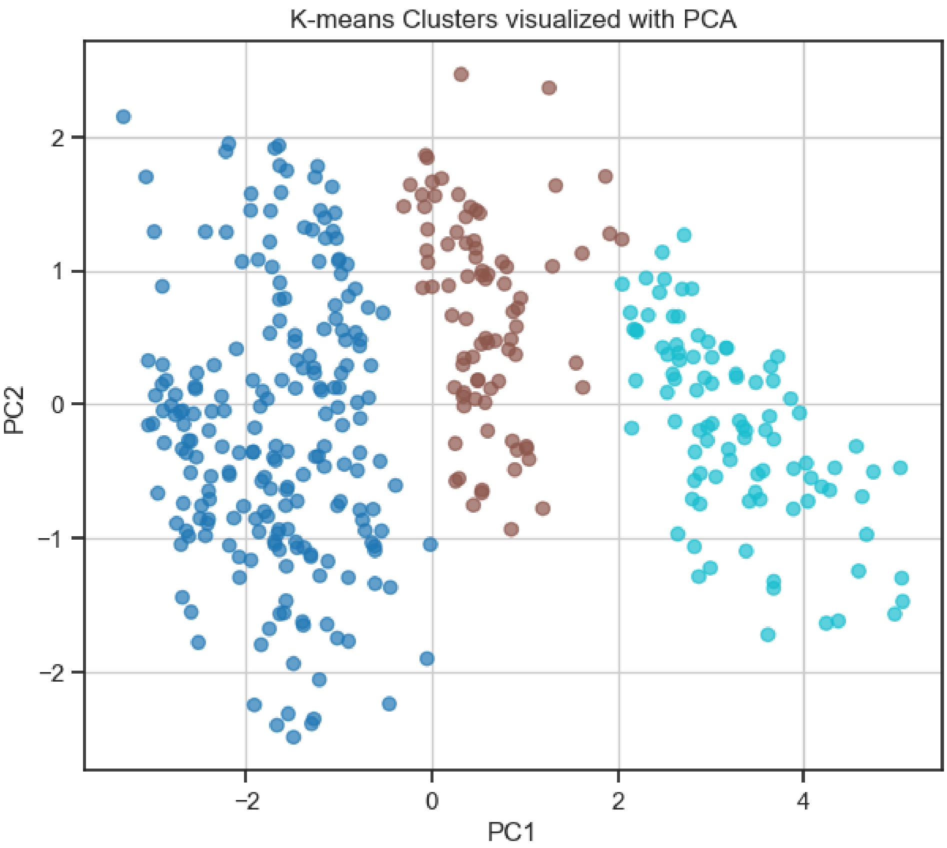


Figure 10: K-means clustering (k = 3) visualized in the first two principal components

The clusters appear as groups of points in the PCA plane, showing that some structure can be detected in the data when projected onto the main components. Although the clusters are not perfectly separated, certain regions of the PCA space correspond preferentially to specific ranges of engine size, weight or model year.

9.2 PCA Projection Colored by Different Features

To further interpret the structure in the PCA space, the same two-dimensional projection is colored by different features such as fuel efficiency, weight, horsepower and number of cylinders.

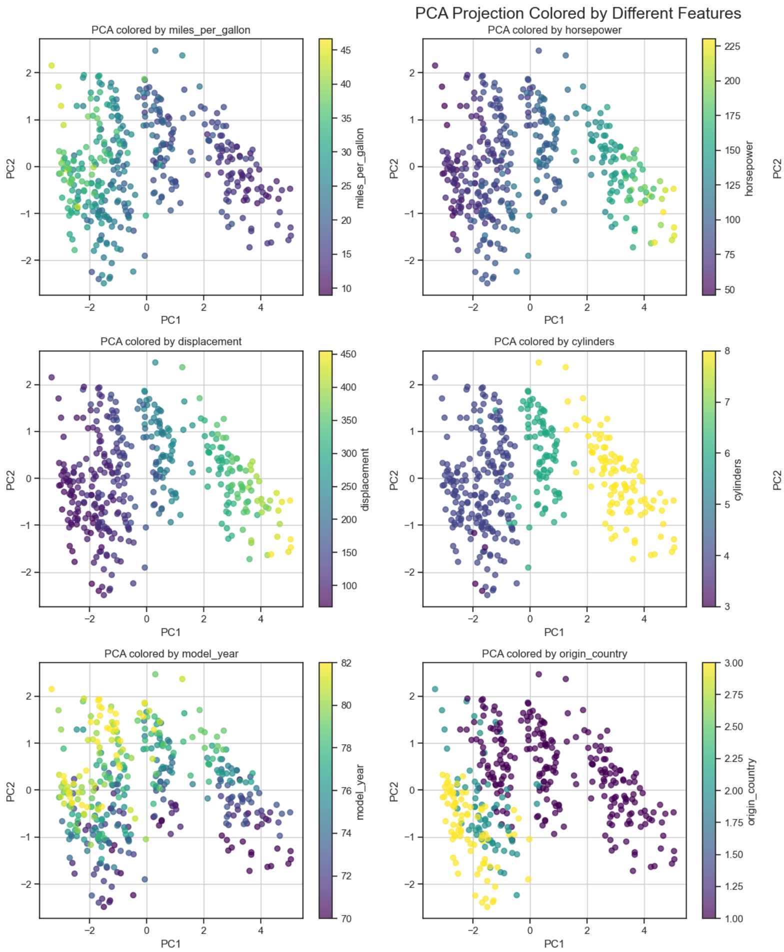


Figure 11: PCA projection colored by different quantitative features

When coloring by `miles_per_gallon`, a smooth gradient appears along one direction of the PCA space, which is consistent with the strong relationships between fuel efficiency and the engine-related variables contributing heavily to PC1. Coloring by `weight_lbs` or `horsepower` reveals similar smooth gradients, indicating that heavier and more powerful cars tend to occupy specific regions of the PCA plane. However, these gradients remain continuous and do not lead to clearly separated groups.

In contrast, when the same PCA projection is colored by `cylinders` and compared with the K-means clustering results for $k = 3$, the three clusters align fairly well with different cylinder configurations. In other words, for $k = 3$, the combination of PCA and K-means essentially recovers groups that can be interpreted mainly in terms of the number of cylinders. Beyond this, the PCA projection colored by other features does not reveal additional clearly separated, easily interpretable clusters, but rather smooth variations along the principal component axes.

10 Conclusion

In this project, we analysed an automobiles dataset from the 1970s–1980s to understand how vehicle characteristics relate to fuel efficiency. The data showed a large variability between cars, clear differences between regions of origin, and strong departures from normality for most raw variables, especially for engine size and weight.

On the modelling side, simple univariate models based only on horsepower captured part of the variability in fuel efficiency, but multivariate models using additional features (weight, displacement, cylinders, model year, origin) performed much better. Among these, MLP models generally achieved the lowest RMSE and highest R^2, while Ridge regression tended to produce residuals that were closer to a normal distribution. PCA confirmed the strong redundancy between engine-related variables and showed that three principal components already summarize most of the information. Using these components slightly degraded Ridge performance but helped the MLP reach its best results, thanks to a more compact and orthogonal representation.

Clustering with K-means in the PCA space revealed some structure, but with $k = 3$ the clusters essentially reflect different numbers of cylinders rather than entirely new, clearly separated groups. This suggests that, in this dataset, unsupervised structure is largely driven by engine configuration.

Finally, all models were evaluated using a single train/test split. In a more advanced setting, using cross-validation (for example k-fold) would provide more reliable estimates of predictive performance and more robust hyperparameter choices, especially for regularized models and neural networks.

Overall, combining descriptive analysis, regression, PCA and clustering gives a coherent picture of how engine size, weight, model year and origin jointly influence fuel efficiency in this historical dataset, and illustrates the respective strengths and limitations of linear and non-linear approaches.